

# NSF Grant Proposal

Directorate for Biological Sciences

**Proposal Number:** BIO-2024-32209

**Submission Date:** November 1, 2024

## Project Information

**Project Title:** Catalytic Conversion of Agricultural Waste to Sustainable Aviation Fuel

**Principal Investigator:** Dr. Amara Okafor

**Institution:** University of Idaho

**Department:** Department of Chemistry

**Requested Amount:** \$410,000

**Project Period:** January 1, 2026 - December 31, 2028

**Sponsoring Agency:** National Science Foundation

**Research Area:** Chemistry

## Project Abstract

Idaho produces over 8 million tons of agricultural residue annually, most of which is burned or left to decompose. This project develops novel heterogeneous catalysts for the thermochemical conversion of wheat straw and potato waste into drop-in sustainable aviation fuel (SAF). We will synthesize and characterize bifunctional zeolite catalysts, optimize pyrolysis and hydrodeoxygenation conditions, and assess the techno-economic viability of a regional SAF production facility. The work addresses DOE and FAA targets for 3 billion gallons of SAF by 2030 while creating economic opportunity for Idaho's agricultural sector.